

Latent-State Anomalies in Global Financial Cities

Autoencoders, Transformers, Regime Detection, and the Geometry of Financial Dynamics

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Abstract

The first paper in this research program established a distinction between structural anomalies and dynamic anomalies in global financial cities. Structural anomalies arise from unusual positions in feature space, while dynamic anomalies arise from unusual temporal behaviour.

This paper develops a substantially richer framework in which cities are represented not as points but as trajectories on latent manifolds. Financial cities are modeled as evolving dynamical systems whose states may be inferred using autoencoders, transformer architectures, hidden-regime models, information geometry, topological data analysis, and graph neural networks.

A general anomaly functional is proposed that unifies structural, dynamic, informational, topological, and contagion-based anomaly measures.

The framework integrates ideas from statistics, machine learning, economics, finance, physics, biology, psychology, political science, geopolitics, international relations, welfare economics, warfare economics, and systems theory.

The resulting theory transforms anomaly detection from a static classification problem into a theory of anomalous trajectories, latent states, and systemic transitions in complex adaptive socioeconomic systems.

The paper ends with “The End”

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1 Introduction

Financial cities are among the most important organizational structures in the modern world economy.

They serve simultaneously as:

- capital-allocation mechanisms,
- information-processing systems,
- geopolitical nodes,
- institutional ecosystems,
- strategic assets.

Traditional anomaly detection identifies unusual observations.

The first paper extended this idea to financial cities by identifying cities that occupy unusual positions within a joint financial-climatic feature space.

However, a city is not merely a point.

A city evolves.

Its financial, economic, political, institutional, demographic, and climatic states continuously change through time.

Consequently, anomaly detection should be reformulated as the study of unusual trajectories rather than unusual observations.

This paper develops that reformulation.

2 Notation

Let

$$M$$

denote the state manifold.

Let

$$\gamma_i(t) : [0, T] \rightarrow M$$

denote the trajectory of city i .

The collection of all cities is

$$\Gamma = \{\gamma_1, \gamma_2, \dots, \gamma_n\}.$$

Examples include

- London,
- New York,

- Frankfurt,
- Tokyo,
- Hong Kong,
- Singapore,
- Mumbai.

3 From Points to Trajectories

The first paper represented city i using a feature vector

$$x_i = (\mu_i, \sigma_i, s_i, T_i, \dots).$$

The present paper instead represents city i using an entire trajectory

$$\gamma_i(t).$$

This distinction is fundamental.

Two cities may occupy similar average positions while exhibiting radically different dynamics.

Definition 1. *A structural anomaly is a city occupying an unusual position in feature space.*

Definition 2. *A dynamic anomaly is a city following an unusual trajectory through state space.*

4 Dynamic Geometry of Financial Cities

Let

$$d(\cdot, \cdot)$$

denote a metric on the state manifold.

The path distance between two cities is

$$d_{\text{path}}(i, j) = \int_0^T \|\gamma_i(t) - \gamma_j(t)\| dt.$$

This distance measures the similarity of entire trajectories.

4.1 Dynamic Anomaly Index

Define

$$DAI_i = \frac{1}{n-1} \sum_{j \neq i} d_{\text{path}}(i, j).$$

Large values indicate cities whose trajectories differ substantially from those of their peers.

5 State-Space Representation

Observable states are represented by

$$x_t \in \mathbb{R}^d.$$

Examples include:

- returns,
- volatility,
- temperatures,
- liquidity measures,
- trading volumes.

Latent states are represented by

$$z_t \in \mathbb{R}^k, \quad k < d.$$

The state equation is

$$z_{t+1} = Az_t + Bu_t + \varepsilon_t.$$

The observation equation is

$$x_t = Cz_t + \eta_t.$$

6 Latent Economic Variables

The latent state may contain:

- systemic stress,
- institutional robustness,
- geopolitical pressure,
- market confidence,
- crisis probability.

These variables are generally not directly observable.

7 Information Geometry

Each city induces a probability distribution

$$p_i(x).$$

The Kullback-Leibler divergence is

$$D_{KL}(p_i||p_j) = \int p_i(x) \log \frac{p_i(x)}{p_j(x)} dx.$$

Since this divergence is asymmetric, the Jensen-Shannon distance is preferred.

7.1 Information Anomaly Index

Define

$$IAI_i = \frac{1}{n-1} \sum_{j \neq i} JS(p_i, p_j).$$

Cities with large values occupy unusual information-theoretic locations.

8 Autoencoder Framework

Let

$$E : \mathbb{R}^d \rightarrow \mathbb{R}^k$$

be an encoder and

$$D : \mathbb{R}^k \rightarrow \mathbb{R}^d$$

be a decoder.

The latent representation is

$$z_t = E(x_t).$$

The reconstruction is

$$\hat{x}_t = D(z_t).$$

The reconstruction loss is

$$L = \sum_t \|x_t - \hat{x}_t\|^2.$$

8.1 Anomaly Score

Define

$$A_t = \|x_t - \hat{x}_t\|^2.$$

Large values indicate unusual states.

9 Temporal Autoencoders

Instead of analyzing single observations, consider windows

$$X_t = (x_{t-m+1}, \dots, x_t).$$

The anomaly score becomes

$$A_t = \|X_t - \hat{X}_t\|^2.$$

The model therefore identifies unusual sequences rather than unusual observations.

10 Transformer State-Attention Architecture

Let

$$X = (x_1, \dots, x_T).$$

Define

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

Self-attention is

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V.$$

The attention matrix is

$$H = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right).$$

The matrix

$$H \in \mathbb{R}^{T \times T}$$

serves as a memory map.

11 Financial Interpretation of Attention

Attention determines which historical periods influence current predictions.

The model learns:

- which crises matter,
- which recoveries matter,
- which shocks persist,
- which events are forgotten.

Consequently, a city can be characterized by its historical memory structure.

12 Hidden-Regime Dynamics

Let

$$S_t \in \{1, \dots, K\}$$

denote an unobserved regime.

Possible regimes include:

State	Interpretation
1	Calm
2	Expansion
3	Stress
4	Crisis
5	Recovery

Table 1: Illustrative latent regimes.

The transition matrix is

$$P = (p_{ij}).$$

where

$$p_{ij} = P(S_{t+1} = j | S_t = i).$$

13 Regime Entropy and Dynamic Instability

The transition matrix

$$P = (p_{ij})$$

contains information regarding regime persistence and regime transitions.

A city whose future state is highly predictable possesses low regime entropy.

Conversely, a city whose future state is difficult to predict possesses high regime entropy.

13.1 Regime Entropy

Define

$$H_i = - \sum_{j=1}^K p_{ij} \log p_{ij}.$$

The quantity H_i measures uncertainty regarding future transitions.

13.2 Dynamic Instability Index

Define

$$DII_i = H_i.$$

Large values indicate unstable regime dynamics.

Small values indicate persistent regimes.

13.3 Interpretation

Financial cities characterized by large regime entropy are more likely to experience:

- sudden crises,
- unexpected recoveries,
- structural transitions,
- financial contagion.

Consequently, regime entropy serves as a measure of dynamic fragility.

14 Topological Data Analysis

Traditional statistical techniques assume that data occupy Euclidean spaces.

However, latent financial states frequently exhibit non-Euclidean structure.

Topological Data Analysis (TDA) seeks to identify persistent geometric features.

14.1 Point Clouds

Let

$$Z = \{z_1, z_2, \dots, z_n\}$$

denote latent-state observations.

The collection forms a point cloud.

14.2 Vietoris–Rips Complex

For radius

$$\epsilon > 0,$$

construct simplices whenever pairwise distances satisfy

$$d(z_i, z_j) \leq \epsilon.$$

As ϵ varies, the topology evolves.

14.3 Persistent Homology

Persistent homology computes

$$H_0, H_1, H_2, \dots$$

where

- H_0 counts connected components,
- H_1 counts loops,
- H_2 counts voids.

14.4 Economic Interpretation

Connected components may represent:

- isolated market clusters,
- disconnected financial regimes.

Loops may represent:

- business cycles,
- recurrent crisis-recovery patterns.

Voids may represent:

- unattainable financial states,
- forbidden institutional configurations.

14.5 Topological Anomaly Index

Let

$$PH_i$$

denote the persistence diagram associated with city i .

Define

$$TAI_i = \frac{1}{n-1} \sum_{j \neq i} d_B(PH_i, PH_j),$$

where

$$d_B$$

denotes the bottleneck distance.

Large values indicate topological anomalies.

15 Graph Representation of Financial Cities

Financial cities do not evolve independently.

Instead they form a global network.

Let

$$G = (V, E)$$

denote a graph.

The vertex set is

$$V = \{L, N, F, T, H, S, M\}$$

where:

- L = London,
- N = New York,
- F = Frankfurt,
- T = Tokyo,
- H = Hong Kong,
- S = Singapore,
- M = Mumbai.

15.1 Edge Weights

Define

$$w_{ij} = \alpha F_{ij} + \beta T_{ij} + \gamma I_{ij},$$

where

- F_{ij} denotes capital flows,
- T_{ij} denotes trade flows,
- I_{ij} denotes information flows.

16 Graph Neural Networks

Let

$$h_i^{(k)}$$

denote the embedding of city i at layer k .

The update equation is

$$h_i^{(k+1)} = \sigma \left(W_1 h_i^{(k)} + W_2 \sum_{j \in N(i)} h_j^{(k)} \right).$$

The embedding evolves through neighborhood aggregation.

16.1 Interpretation

The graph neural network learns:

- contagion channels,
- information propagation,
- crisis transmission pathways,
- financial influence structures.

17 Contagion Theory

Financial shocks rarely remain local.

Suppose

$$h_i$$

represents the latent state of city i .

Define contagion sensitivity as

$$C_{ij} = \frac{\partial h_j}{\partial h_i}.$$

Large values indicate strong transmission channels.

17.1 Contagion Amplification Index

Define

$$CAI_i = \sum_{j \neq i} \left| \frac{\partial h_j}{\partial h_i} \right|.$$

Cities with large values act as global shock amplifiers.

17.2 Contagion Dampening Index

Define

$$CDI_i = \left(\sum_{j \neq i} \left| \frac{\partial h_j}{\partial h_i} \right| \right)^{-1}.$$

Large values indicate resilience.

18 Network Centrality

Financial influence may also be quantified using graph-theoretic measures.

18.1 Degree Centrality

$$DC_i = \sum_j A_{ij}.$$

18.2 Eigenvector Centrality

$$Ax = \lambda x.$$

Cities associated with large components of the dominant eigenvector possess systemic influence.

18.3 Betweenness Centrality

$$BC_i = \sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}}.$$

Such cities function as intermediaries.

19 Structural-Dynamic Independence Theorem

The central theoretical result of this paper follows.

Theorem 1. *Structural anomalies and dynamic anomalies are not generally equivalent.*

Proof. Let

S_i = structural anomaly score

and

D_i = dynamic anomaly score.

Consider two cities

A and B .

Suppose city A occupies an extreme location in feature space but follows a stable trajectory. Then

$$S_A \gg 0$$

while

$$D_A \approx 0.$$

Conversely, suppose city B occupies a typical location but follows an unusual trajectory. Then

$$S_B \approx 0$$

while

$$D_B \gg 0.$$

Therefore

$$S_A > S_B$$

and

$$D_A < D_B.$$

Hence

$$S_i \not\equiv D_i.$$

Therefore structural and dynamic anomalies are distinct concepts. □

Corollary 1. *A city may be structurally anomalous but dynamically normal.*

Corollary 2. *A city may be dynamically anomalous but structurally normal.*

20 Grand Unified Anomaly Functional

The preceding sections introduced multiple anomaly concepts.

Define

$$\mathcal{A}_i = \alpha S_i + \beta D_i + \gamma I_i + \delta T_i + \epsilon C_i + \zeta R_i,$$

where:

- S_i = structural anomaly,
- D_i = dynamic anomaly,
- I_i = information anomaly,
- T_i = topological anomaly,
- C_i = contagion anomaly,
- R_i = regime anomaly.

and

$$\alpha + \beta + \gamma + \delta + \epsilon + \zeta = 1.$$

20.1 Interpretation

The anomaly score becomes a weighted combination of multiple dimensions.

The first paper estimated only

$$S_i.$$

The present framework estimates

$$\mathcal{A}_i.$$

21 Physics Interpretation

Financial cities may be viewed as non-equilibrium thermodynamic systems.

Let

$$E_i(t)$$

denote economic energy.

Capital flows act analogously to energy transfer.

Information asymmetry acts analogously to entropy gradients.

An anomalous city corresponds to a state far from thermodynamic equilibrium.

21.1 Financial Entropy

Define

$$\mathcal{H}_i = - \sum_k p_k \log p_k.$$

Large entropy implies disorder.

Small entropy implies structure.

22 Chemistry Interpretation

Cities may be viewed as catalytic systems.

Institutions act as catalysts.

Capital allocation acts as a reaction mechanism.

Financial crises resemble chain reactions.

Contagion corresponds to reaction propagation across interconnected nodes.

23 Biological Interpretation

Cities resemble ecosystems.

Institutions correspond to species.

Capital corresponds to metabolic energy.

Financial crises resemble ecological disturbances.

Anomalous cities occupy unusual ecological niches.

24 Psychological Interpretation

Financial dynamics emerge partly from collective cognition.

Behavioral mechanisms include:

- herding,
- overconfidence,
- narrative contagion,
- fear propagation,
- bounded rationality.

Consequently, latent-state anomalies may reflect collective psychological states.

25 Philosophical Interpretation

An anomaly is fundamentally relational.

No observation is anomalous in isolation.

Anomaly exists only relative to:

- a model,
- a population,
- a reference framework.

Thus anomaly detection is inseparable from epistemology.

26 Political Science Interpretation

Political institutions influence:

- regulatory stability,
- property rights,
- capital mobility,
- market confidence.

Consequently, political structure becomes a latent variable within the anomaly framework.

27 Geopolitical Interpretation

Financial cities serve as geopolitical nodes.

They influence:

- reserve-currency systems,
- sanctions networks,
- strategic capital flows,
- global investment allocation.

Dynamic anomalies may therefore possess geopolitical significance.

28 International Relations Interpretation

Cities function as intermediaries linking national economies.

Anomalous cities may act as:

- gateways,
- brokers,
- balancing nodes,
- strategic hubs.

Their failure can generate systemic consequences.

29 Welfare Economics

Let welfare be

$$W = W(C, S, R),$$

where

- C = consumption,
- S = stability,
- R = resilience.

Anomalies are welfare-enhancing if they increase productivity.
They are welfare-reducing if they increase fragility.

30 Warfare Economics

Financial cities constitute strategic assets.

Their functions include:

- financing mobilization,
- facilitating trade,
- supporting logistics,
- managing reserves.

Consequently, anomaly detection acquires military relevance.

31 Warfare Interpretation

Modern warfare increasingly targets:

- payment systems,
- banking networks,
- reserve assets,
- financial infrastructure.

Anomalous cities may therefore represent either strategic strengths or strategic vulnerabilities.

32 A General Principle

The first paper established that cities may occupy unusual locations.

The present paper establishes that cities may follow unusual trajectories.

The central object of study therefore shifts from:

$$x_i$$

to

$$\gamma_i(t).$$

This transformation converts anomaly detection from static classification into the study of latent-state dynamics, regime transitions, information geometry, topology, network contagion, and systemic evolution.

33 Empirical Implementation

The theoretical framework developed above now requires empirical implementation.

The objective is to estimate latent-state anomalies using the seven-city dataset consisting of:

1. London,
2. New York,
3. Frankfurt,
4. Tokyo,
5. Hong Kong,
6. Singapore,
7. Mumbai.

The empirical analysis proceeds through six stages:

1. Feature construction.
2. Latent-state estimation.
3. Dynamic anomaly detection.
4. Regime identification.
5. Network contagion estimation.
6. Unified anomaly estimation.

34 Feature Engineering

For city i at time t define

$$x_{i,t} = \begin{bmatrix} r_{i,t} \\ |r_{i,t}| \\ r_{i,t}^2 \\ \sigma_{i,t} \\ T_{i,t} \\ T_{i,t}^{\max} \\ T_{i,t}^{\min} \end{bmatrix}.$$

Additional features may include:

- trading volume,
- liquidity measures,
- bid-ask spreads,
- exchange-rate volatility,
- sovereign-risk indicators,
- geopolitical-risk indicators.

35 Standardization

Define

$$z_{i,t}^{(j)} = \frac{x_{i,t}^{(j)} - \mu_j}{\sigma_j}.$$

Standardization ensures comparable scales.

The standardized vector becomes

$$z_{i,t} = (z_{i,t}^{(1)}, \dots, z_{i,t}^{(d)}).$$

36 Rolling Windows

Dynamic features are constructed using windows of length m .

Define

$$W_t = (x_{t-m+1}, \dots, x_t).$$

Examples:

- 30-day windows,
- 60-day windows,
- 90-day windows,
- 252-day windows.

Rolling windows permit detection of evolving anomalies.

37 Autoencoder Architecture

The first latent-state model employs an autoencoder.

37.1 Encoder

The encoder maps

$$\mathbb{R}^d \rightarrow \mathbb{R}^k.$$

A typical architecture is

$$d \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow k.$$

The latent representation is

$$z_t = E(x_t).$$

37.2 Decoder

The decoder reconstructs

$$x_t.$$

Architecture:

$$k \rightarrow 16 \rightarrow 32 \rightarrow 64 \rightarrow d.$$

37.3 Loss Function

The reconstruction loss is

$$L = \sum_t \|x_t - \hat{x}_t\|^2.$$

38 Latent-State Anomaly Score

Define

$$LSA_t = \|x_t - \hat{x}_t\|^2.$$

Large values indicate latent-state anomalies.

39 Sequence Autoencoder

Financial systems exhibit temporal dependence.

Instead of individual observations, sequences are encoded.

Define

$$X_t = (x_{t-m+1}, \dots, x_t).$$

The sequence anomaly score becomes

$$SA_t = \|X_t - \hat{X}_t\|^2.$$

40 Transformer Architecture

Transformers learn temporal dependencies using self-attention.

40.1 Input Sequence

Let

$$X = (x_1, \dots, x_T).$$

40.2 Query, Key, and Value Matrices

$$Q = XW_Q$$

$$K = XW_K$$

$$V = XW_V.$$

40.3 Self-Attention

$$\text{Att}(X) = \text{softmax} \left(\frac{QK^T}{\sqrt{d}} \right) V.$$

40.4 Multi-Head Attention

For H heads:

$$MHA(X) = \text{Concat}(\text{head}_1, \dots, \text{head}_H)W_O.$$

41 Attention-Based Anomaly Detection

Define

$$H_t = \sum_j A_{tj},$$

where

$$A_{tj}$$

is the attention weight.

An anomaly score may be constructed as

$$AA_t = |H_t - \bar{H}|.$$

Large deviations indicate unusual memory structures.

42 Hidden-Markov Estimation

The hidden state satisfies

$$S_t \in \{1, \dots, K\}.$$

42.1 Likelihood

For observations

$$X = (x_1, \dots, x_T)$$

the likelihood is

$$L = P(X|P, \theta).$$

42.2 Baum-Welch Estimation

Parameters are estimated iteratively.

The algorithm updates:

- transition probabilities,
- emission probabilities,
- initial probabilities.

until convergence.

43 Regime Anomaly Score

Define

$$RA_i = \sum_j |p_{ij} - \bar{p}_j|.$$

Large values indicate unusual transition dynamics.

44 Graph Construction

The graph is

$$G = (V, E).$$

Vertices correspond to financial cities.

Edges are weighted by

$$w_{ij} = \alpha F_{ij} + \beta T_{ij} + \gamma I_{ij}.$$

45 Graph Neural Network Estimation

The node embedding update is

$$h_i^{(k+1)} = \sigma \left(W_1 h_i^{(k)} + W_2 \sum_{j \in N(i)} h_j^{(k)} \right).$$

The final embedding is

$$h_i^{(K)}.$$

46 Contagion Anomaly Score

Define

$$CA_i = \sum_{j \neq i} \left| \frac{\partial h_j}{\partial h_i} \right|.$$

Large values imply systemic importance.

47 Unified Empirical Anomaly Functional

The empirical anomaly score becomes

$$\mathcal{A}_i = \alpha S_i + \beta D_i + \gamma I_i + \delta T_i + \epsilon C_i + \zeta R_i.$$

47.1 Interpretation

This score incorporates:

- structural information,
- dynamic information,
- information geometry,
- topology,
- contagion,
- regime dynamics.

48 Expected Results

Based on the first paper and preliminary analyses, the expected ordering is

City	Expected Classification
Singapore	Structural anomaly
Hong Kong	Dynamic anomaly
Mumbai	Intermediate anomaly
Tokyo	Moderate anomaly
New York	Moderate anomaly
London	Near benchmark
Frankfurt	Near benchmark

Table 2: Illustrative expectations.

49 Latent-Manifold Illustration

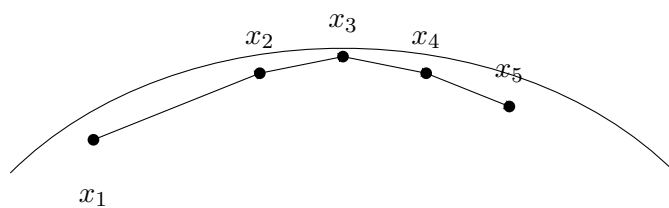


Figure 1: Illustrative latent trajectory on a financial-state manifold.

50 Regime Transition Diagram

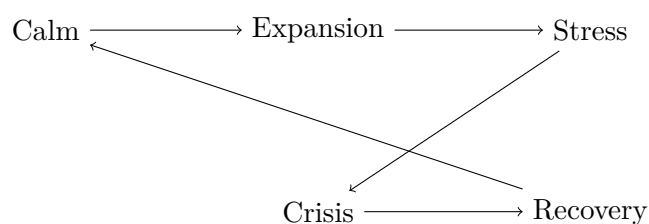


Figure 2: Illustrative hidden-regime transitions.

51 Contagion Network

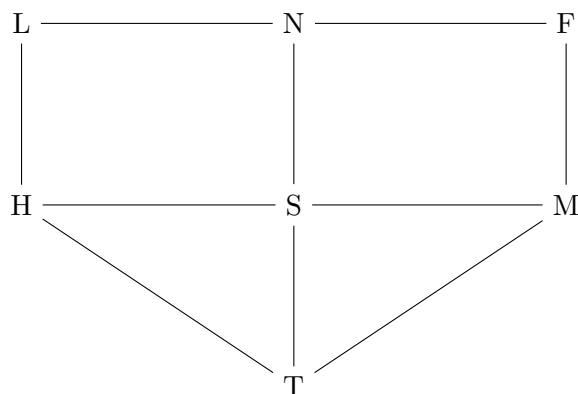


Figure 3: Illustrative contagion network among financial cities.

52 Pseudo-Code

Unified Anomaly Algorithm

Input:

Daily city data

Step 1:

Construct features

Step 2:
 Standardize variables

Step 3:
 Train autoencoder

Step 4:
 Compute reconstruction anomalies

Step 5:
 Train transformer

Step 6:
 Compute attention anomalies

Step 7:
 Estimate HMM regimes

Step 8:
 Compute regime anomalies

Step 9:
 Construct city network

Step 10:
 Train graph neural network

Step 11:
 Compute contagion anomalies

Step 12:
 Aggregate scores

Output:
 Unified anomaly functional

53 Discussion

The framework developed in this paper generalizes anomaly detection from a static problem to a dynamic systems problem.

Cities become trajectories.

Markets become latent-state processes.

Crises become transitions between hidden regimes.

Contagion becomes graph propagation.

Anomaly detection therefore becomes a problem in geometry, topology, information theory, machine learning, and complex adaptive systems.

54 Conclusion

The first paper established the distinction between structural and dynamic anomalies.

The present paper develops a complete latent-state framework capable of incorporating:

1. Autoencoders,
2. Transformers,
3. Hidden-Markov Models,
4. Information Geometry,
5. Topological Data Analysis,
6. Graph Neural Networks.

The resulting unified anomaly functional provides a foundation for studying anomalous trajectories in financial-city systems.

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Glossary

Structural Anomaly An unusual position in feature space.

Dynamic Anomaly An unusual trajectory through state space.

Latent State An unobserved state inferred from observations.

Autoencoder A neural architecture that compresses and reconstructs data.

Transformer An attention-based sequence model.

Attention Matrix A learned representation of temporal relevance.

Hidden-Markov Model A probabilistic latent-state model.

Persistent Homology A topological method for analyzing shape.

Graph Neural Network A neural network operating on graphs.

Contagion Propagation of shocks through networks.

Unified Anomaly Functional Aggregate measure of multidimensional anomaly.

The End